

Name: K RAJ BHARATH K RAJ BHARATH | Reg. No: RA2511003011548

Unit – 4 : Session – 6 : SLO – 1 | Expression Tree

■ I. Fill in the Blanks

- An EXPRESSION TREE is a binary tree representing arithmetic expressions.
- The LEAVES of an expression tree are operands.
- The INTERNAL NODES of an expression tree are operators.
- The INORDER traversal of an expression tree gives the infix expression.
- The POSTORDER traversal of an expression tree gives the postfix expression.

■ II. Expression Tree for: $a + (b * c) + d * (e + f)$

- Fully parenthesised: $(a + ((b * c) + (d * (e + f))))$
- Tree structure (indentation shows children):
 - +
 - /\
 - a +
 - /\
 - * *
 - /\ /\
 - b c d +
 - /\
 - e f
- Root = + (the last operation evaluated).
- Inorder (infix): $a + b * c + d * e + f$
- Postorder (postfix): $a b c * d e f + * + +$